Avi Sharma

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Portfolio: av1sharma.github.io | LinkedIn: linkedin.com/in/avi-sharma-aa5a73223 | GitHub: github.com/Av1Sharma

Education

Cornell University Expected May 2029
Bachelor of Arts in Computer Science *Ithaca, NY*

Experience

IT Help Desk Intern Jun. – Aug. 2026
Prince William County Public Schools *Prince William County, VA*

- Maintained school-issued student and staff laptops and installed software updates in a district technology environment managing 146,000+ devices.
- Troubleshot hardware, software, and network issues; supported device configuration, inventory, and deployment preparation with district IT staff.

Lead Programmer & Overall Team Lead 2024 – 2026
1885 COMET Robotics *Haymarket, VA*

- Led software for a 40+ member FRC team; authored 9 merged PRs in the public 2025/2026 robot repositories and contributed to the team's first District Championship qualification.
- Implemented Java/WPILib PID heading correction for a 4-module swerve drivetrain using gyro-based pose feedback and 20 ms chassis-speed discretization.
- Integrated distance-interpolated flywheel RPM and hood-angle setpoints; gated firing on a configured 70 RPM tolerance. Added MapleSim simulation and AdvantageKit logging for debugging.

Selected Projects & Open Source

ClearPath | *JavaScript, Node.js, Leaflet, WebSockets*
GitHub: github.com/rzli2106/bigredhacks2026 | Live: www.clearpath.wiki/mobile

- Co-built a deployed campus navigation prototype using binary min-heap Dijkstra search, KD-tree hazard matching, and exponentially decaying edge penalties; authored 15 merged PRs in the shared repository.
- Fixed overlapping GPS/route requests with AbortController and generation guards; serialized native health-sharing startup, polling, and shutdown to suppress stale callbacks and concurrent reads.
- Restored navigation lifecycle handling after a merge caused 31 failing tests; added a 3-second passed-hazard prompt and WebSocket disconnect/recovery status without discarding the selected route.
- Updated GitHub Actions web/native workflows; successful CI runs cover web tests/builds and Android/iOS compilation. Added regression tests for authentication expiry, queue overflow, cancellation, and retry.

Lap Lab | *React, TypeScript, Vite, SVG, OpenF1*
GitHub: github.com/Av1Sharma/F1-Telemetry-App | Live: av1sharma.github.io/lap-lab/

- Deployed a telemetry explorer with a bundled Monza demo, 1950–2026 race-results archive, lap replay, sector comparisons, and 7-column CSV export.
- Aligned traces using trapezoidal speed integration, circuit-length normalization, binary search, and linear interpolation; preserved discrete brake/gear values and rejected incomplete recordings.
- Serialized historical API requests with 2.1-second spacing, an 80-entry cache, 25-second timeouts, and cancellation; added Node.js tests for interpolation, CSV formatting, and telemetry normalization.

Leadership & Activities

Cornell AppDev | SQL Backend Development Sep. 2026 – Present
Developing SQL backend skills through database queries and API integration.

Cornell Hack4Impact | Mentee Sep. 2026 – Present
Participating in freshman mentorship focused on software development and technical growth.

Technical Skills

Languages: Python, Java, C++, Swift, JavaScript, TypeScript, SQL, HTML/CSS
Frameworks / Libraries: React, Node.js, Leaflet, WPILib, MapleSim, OpenCV, SQLite
Tools: Git/GitHub, GitHub Actions, Vite, AdvantageKit, VS Code, Xcode, Arduino